

# Francisco Rodríguez

Data & AI Engineer | LLMs | Agentic Systems | Data Engineering

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## Professional Summary

Engineer with a hybrid profile across **data engineering and AI**, a scientific background and a **product-oriented mindset**. I bring **end-to-end vision** from data to production, analytical rigor and effective communication with technical and business stakeholders, turning strategic needs into **robust, governable and impact-measurable solutions**.

## Professional Experience

### Gran Telescopio de Canarias (GRANTECAN), Data & AI Engineer

07/2025 - Present

- **Member of the organization's pioneer on-premise AI team:** led the study and validation of **compliance with the EU AI Act and AESIA**, shaping the **privacy and governance framework** on top of which the PoCs, MVPs and production systems described below were built.
- **Co-developed an agentic talk-to-your-data system** (speech-to-text, LangGraph and RAG over Deltalake) that lets non-technical users query in natural language with no schema or variable-name knowledge and **receive graphical answers**, democratizing data access across the organization.
- **Designed an agentic assistant (RAG over local LLMs)** for the assessment and drafting of **technical and administrative tender specifications, reducing manual workload by ~50%** for the technical-administrative staff and bringing corporate context into the documentation process.
- **Built PySpark ETL pipelines on the corporate Deltalake:** ingested new sources to **break information silos** and promoted data from **raw to silver and gold layers** (medallion architecture), accelerating on-demand telemetry access and offloading the data team for both technical and non-technical users.
- **Deployed the local inference infrastructure on a GPU cluster** with **vLLM, SLURM and RHEL**, technical backbone for the agentic systems above and a key enabler of the on-premise approach.
- **Led the build of a SOP and playbooks platform** as a **procedure sequencer-orchestrator for VUCA environments:** rigorous version control and execution traceability, removing reliance on tacit knowledge and outdated procedures scattered across emails and documents.
- **Developed a React and Node.js web app for telescope time-usage analytics** for management and operations: asynchronous cross-validation of incident logs and observation reports in a single tool, **replacing manual Excel-based coordination** between teams.

### Cabildo of La Palma, Data Scientist

01/2024 - 01/2025

- **Led EDA with Python and R across 5+ air-quality datasets** in Puerto Naos, uncovering a **Pearson correlation of ~0.6 between coastal CO2 levels and tidal cycles**; the finding drove the Emergency Department's post-eruption habitability protocol review.
- **Designed interactive dashboards** in Power BI, Tableau and ArcGIS Online (including a **Puerto Naos digital twin** used daily by Emergency for volcanic CO2 tracking) and automated weekly Python workflows, **saving ~12 h per month** for non-technical stakeholders.
- **Standardized and integrated a 3M+ record legacy dataset into PostgreSQL**, extending the Cabildo's accessible meteorological archive from 2022 back to 2016 and ensuring continuity for the public portal.

*Previous experience as private tutor in Physics, Chemistry and Mathematics.*

## Education

### MSc in Data Science and Big Data, Open University of Catalonia (UOC)

2025

- **Master's Thesis:** sales forecasting and audience segmentation system for Filarmonía de Madrid with real ticketing data (Random Forest, CNN, RNN, ARIMA, K-Means, DBSCAN, RFM).

### BSc in Physics, University of La Laguna (ULL)

2021

- **Bachelor's Thesis:** photometric calibration of the EMIR instrument (GTC, IAC) using Python and scientific documentation in LaTeX.

## Skills

**Programming:** Python, SQL, Node.js, React, TypeScript, R

**AI & Machine Learning:** LangChain, LangGraph, vLLM, Ollama, RAG, agentic systems, prompt engineering, embeddings, vector databases, NLP, deep learning, PyTorch, TensorFlow, Scikit-learn

**Data Engineering:** PySpark, Deltalake (medallion architecture), ETL, data pipelines, Apache Kafka, Trino, PostgreSQL

**MLOps & DevOps:** Linux, SLURM, Docker, Kubernetes, CI/CD, Git, REST API, Nginx, Apache

**Cloud & Visualization:** AWS, Power BI, Tableau, ArcGIS Online

**Templating & Markup:** Jinja, LaTeX

**Languages:** Spanish (native), English (fluent)